

Detecting Labor Market Shifts Using Big Data Analytics on Large-Scale Job Posting Data

CLUSTER 05

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Introduction & Executive Summary

This project demonstrates an end-to-end Big Data engineering and analytics pipeline designed to detect labor market shifts using a large-scale dataset of LinkedIn job postings. In an era where economic trends change rapidly, traditional economic indicators fail to provide real-time updates due to computational and administrative delays. By leveraging a distributed architecture running on an on-premises Hadoop Distributed File System (HDFS) and Apache Spark framework, this system successfully ingests, cleans, analyzes, and clusters millions of data points. The final pipeline delivers actionable insights into geographic hiring distributions, shifting skill frequencies, and latent role similarities through automated machine learning algorithms.

Problem Statement & Objectives

Problem Statement

Official labor market reports (such as those from the Bureau of Labor Statistics) are typically published with a 1-2 month lag. This reporting gap makes it highly challenging for policymakers, corporations, and economists to respond to rapid employment shifts or sudden industry declines in real time. While millions of online job postings are generated daily containing rich, timely signals about hiring demand, shifting skill requirements, and sectoral changes their sheer volume introduces severe memory bottlenecks. Traditional single-machine tools (like Pandas) cannot process this data efficiently, making distributed Big Data storage and parallel processing technologies absolutely essential.

Project Objectives

Objective 1: Ingest and securely store the large-scale LinkedIn job postings dataset on a $\text{\$4\text{-node}}$ on-premises HDFS cluster.

Objective 2: Perform distributed preprocessing, structural standardization, and automated data cleaning using Apache Spark.

Objective 3: Conduct multi-dimensional exploratory data analysis (EDA) via Spark SQL to uncover top hiring sectors, skill demand trends, geographic distribution, and salary scales.

Objective 4: Develop and execute a distributed K-Means clustering model using Spark MLlib to group job profiles by economic demands and uncover hidden structural shifts.

Objective 5: Persist and export structured result directories back to HDFS for subsequent reporting, dashboard visualization, and downstream analytics.

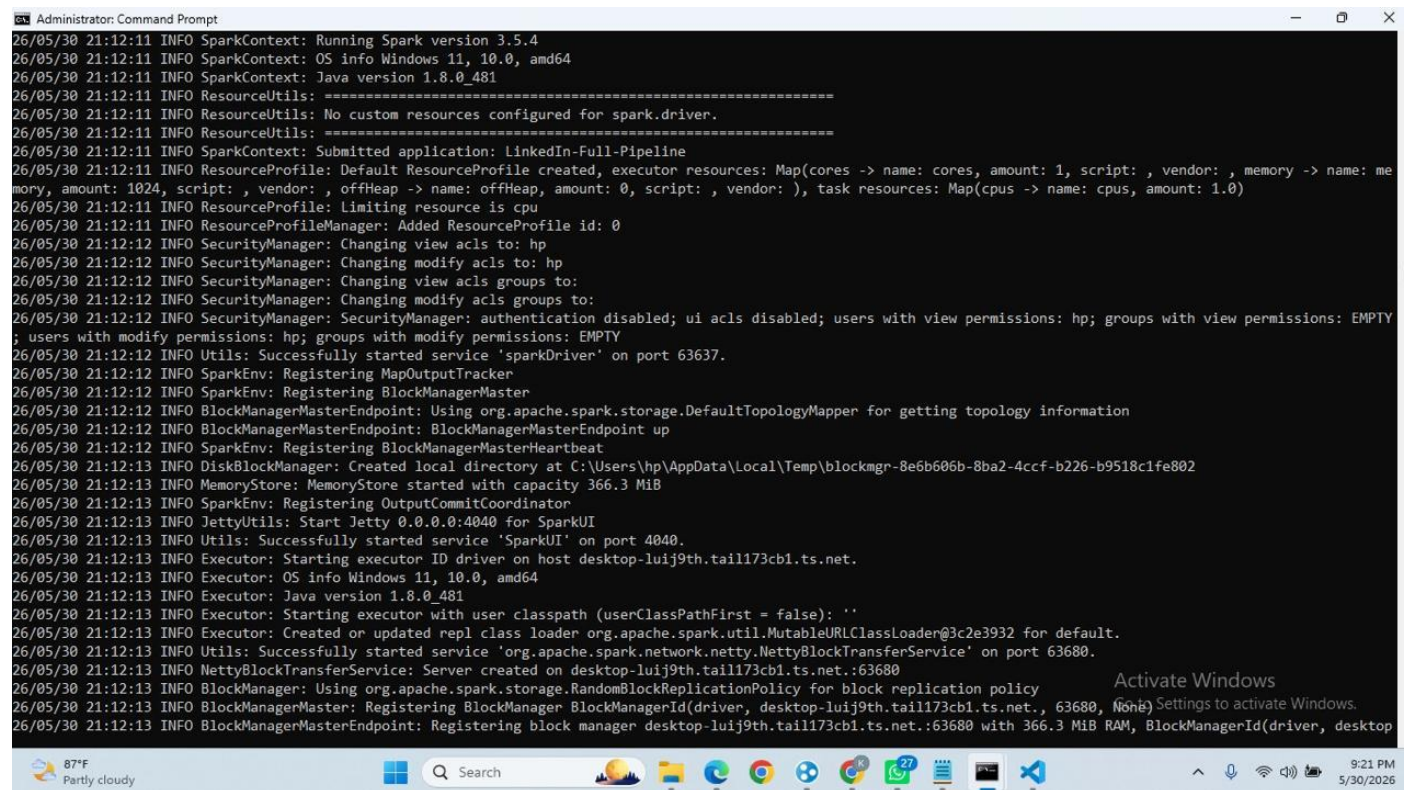
Cluster Application Ingestion

The pipeline script is submitted to the cluster manager, successfully spawning SparkContext under the custom application name LinkedIn-Full-Pipeline. The terminal logs verify driver allocation, environment security hooks, and the deployment of the Spark UI on port 4040.

Command:

```
spark-submit ^ --master yarn ^ --deploy-mode cluster ^ --num-executors 4 ^ --executor-cores 3 ^  
--executor-memory 5g ^ --driver-memory 2g ^ --conf spark.executor.memoryOverhead=1g ^ --  
conf spark.sql.shuffle.partitions=200 ^ --conf spark.default.parallelism=24 ^ --conf  
spark.yarn.stagingDir=/tmp/spark-staging ^ --conf spark.hadoop dfs.client.socket-  
timeout=120000 ^ --conf spark.hadoop dfs.datanode.socket.write.timeout=120000 ^ --conf  
spark.hadoop dfs.replication=2 ^
```

spark_pipeline.py



```
Administrator: Command Prompt
26/05/30 21:12:11 INFO SparkContext: Running Spark version 3.5.4
26/05/30 21:12:11 INFO SparkContext: OS info Windows 11, 10.0, amd64
26/05/30 21:12:11 INFO SparkContext: Java version 1.8.0_481
26/05/30 21:12:11 INFO ResourceUtils: =====
26/05/30 21:12:11 INFO ResourceUtils: No custom resources configured for spark.driver.
26/05/30 21:12:11 INFO ResourceUtils: =====
26/05/30 21:12:11 INFO SparkContext: Submitted application: LinkedIn-Full-Pipeline
26/05/30 21:12:11 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory -> name: me
memory, amount: 1024, script: , vendor: , offHeap -> name: offHeap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
26/05/30 21:12:11 INFO ResourceProfile: Limiting resource is cpu
26/05/30 21:12:11 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/05/30 21:12:12 INFO SecurityManager: Changing view acls to: hp
26/05/30 21:12:12 INFO SecurityManager: Changing modify acls to: hp
26/05/30 21:12:12 INFO SecurityManager: Changing view acls groups to:
26/05/30 21:12:12 INFO SecurityManager: Changing modify acls groups to:
26/05/30 21:12:12 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: hp; groups with view permissions: EMPTY
; users with modify permissions: hp; groups with modify permissions: EMPTY
26/05/30 21:12:12 INFO Utils: Successfully started service 'sparkDriver' on port 63637.
26/05/30 21:12:12 INFO SparkEnv: Registering MapOutputTracker
26/05/30 21:12:12 INFO SparkEnv: Registering BlockManagerMaster
26/05/30 21:12:12 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
26/05/30 21:12:12 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
26/05/30 21:12:12 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/05/30 21:12:13 INFO DiskBlockManager: Created local directory at C:\Users\hp\AppData\Local\Temp\blockmgr-8e6b606b-8ba2-4ccf-b226-b9518c1fe802
26/05/30 21:12:13 INFO MemoryStore: MemoryStore started with capacity 366.3 MiB
26/05/30 21:12:13 INFO SparkEnv: Registering OutputCommitCoordinator
26/05/30 21:12:13 INFO JettyUtils: Start Jetty 0.0.0.0:4040 for SparkUI
26/05/30 21:12:13 INFO Utils: Successfully started service 'SparkUI' on port 4040.
26/05/30 21:12:13 INFO Executor: Starting executor ID driver on host desktop-luij9th.tail173cb1.ts.net.
26/05/30 21:12:13 INFO Executor: OS info Windows 11, 10.0, amd64
26/05/30 21:12:13 INFO Executor: Java version 1.8.0_481
26/05/30 21:12:13 INFO Executor: Starting executor with user classpath (userClassPathFirst = false): ''
26/05/30 21:12:13 INFO Executor: Created or updated repl class loader org.apache.spark.util.MutableURLClassLoader@3c2e3932 for default.
26/05/30 21:12:13 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 63680.
26/05/30 21:12:13 INFO NettyBlockTransferService: Server created on desktop-luij9th.tail173cb1.ts.net.:63680
26/05/30 21:12:13 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/05/30 21:12:13 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, desktop-luij9th.tail173cb1.ts.net., 63680, 63680) Settings to activate Windows.
26/05/30 21:12:13 INFO BlockManagerMasterEndpoint: Registering block manager desktop-luij9th.tail173cb1.ts.net.:63680 with 366.3 MiB RAM, BlockManagerId(driver, desktop
```

```
Administrator: Command Prompt
26/05/30 21:12:13 INFO Executor: Java version 1.8.0_481
26/05/30 21:12:13 INFO Executor: Starting executor with user classpath (userClassPathFirst = false): ''
26/05/30 21:12:13 INFO Executor: Created or updated repl class loader org.apache.spark.util.MutableURLClassLoader@3c2e3932 for default.
26/05/30 21:12:13 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 63680.
26/05/30 21:12:13 INFO NettyBlockTransferService: Server created on desktop-luij9th.tail173cb1.ts.net.:63680
26/05/30 21:12:13 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/05/30 21:12:13 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, desktop-luij9th.tail173cb1.ts.net., 63680, None)
26/05/30 21:12:13 INFO BlockManagerMasterEndpoint: Registering block manager desktop-luij9th.tail173cb1.ts.net.:63680 with 366.3 MIB RAM, BlockManagerId(driver, desktop-luij9th.tail173cb1.ts.net., 63680, None)
26/05/30 21:12:13 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, desktop-luij9th.tail173cb1.ts.net., 63680, None)
26/05/30 21:12:13 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, desktop-luij9th.tail173cb1.ts.net., 63680, None)

LinkedIn Jobsb- Full Spark Pipeline

=====
STAGE 1 - PREPROCESSING
=====
Reading raw CSV
Raw rows: 9,549
Parsing dates and building temporal features
Engineering salary features
Exploding skills ...
Writing clean Parquet (partitioned by sector / post_year) ...
26/05/30 21:12:32 WARN SparkStringUtils: Truncated the string representation of a plan since it was too large. This behavior can be adjusted by setting 'spark.sql.debug.maxToStringFields'.
Writing exploded skills Parquet

Stage 1 done - 9,549 clean rows, 60,796 skill rows

=====
STAGE 2 - EDA AGGREGATIONS
=====
[A] Sector demand map
sector_demand_geo
[B] Sector monthly trend
sector_monthly_trend

Activate Windows
Go to Settings to activate Windows.
```

```
Administrator: Command Prompt - spark-submit --master yarn --deploy-mode cluster --num-executors 4 --executor-cores 3 --executor-memory 5g --driver-memory 2g --conf spark.executor.memoryOverhead=1g --conf spar...
Parsing dates and building temporal features
Engineering salary features
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Writing exploded skills Parquet

Stage 1 done - 9,549 clean rows, 60,796 skill rows

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STAGE 2 - EDA AGGREGATIONS
=====
[A] Sector demand map
sector_demand_geo
[B] Sector monthly trend
sector_monthly_trend
[C] Skill frequency
skill_freq_overall
[D] Skill YoY trend
skill_trend_yoy
[E] Geo heatmap
geo_heatmap
[F] Salary distribution
salary_distribution
[G] Remote trend
remote_trend

Stage 2 done - 7 result tables written

=====
STAGE 3 - ML CLUSTERING
=====
Rows for clustering: 9,549
Fitting TF-IDF - PCA - Scaler pipeline
26/05/30 21:15:06 WARN InstanceBuilder: Failed to load implementation from:dev.ludovic.netlib.lapack.JNILAPACK
Feature matrix cached on workers.
Tuning K (k=4..12) via silhouette score

Activate Windows
Go to Settings to activate Windows.
```

```
Administrator: Command Prompt - spark-submit --master yarn --deploy-mode cluster --num-executors 4 --executor-cores 3 --executor-memory 5g --driver-memory 2g --conf spark.executor.memoryOverhead=1g --conf spar...
[B] Sector monthly trend
    sector_monthly_trend
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Feature matrix cached on workers.
Tuning K (k=4..12) via silhouette score
26/05/30 21:15:13 WARN InstanceBuilder: Failed to load implementation from:dev.ludovic.netlib.blas.JNIBLAS
k= 4 silhouette=0.1386
k= 5 silhouette=0.1144
k= 6 silhouette=0.0982
k= 7 silhouette=0.0777
k= 8 silhouette=0.0824
k= 9 silhouette=0.0401
k=10 silhouette=0.0348
k=11 silhouette=0.0421
k=12 silhouette=0.0072

Best k=8 silhouette=0.0824
Writing cluster assignments
Computing cluster profiles
Computing 2-D PCA scatter data

Activate Windows
Go to Settings to activate Windows.
```

Distributed Data Preprocessing (Stage 1)

In Stage 1, Spark reads the raw text, cleans missing fields, parses salaries, and breaks down multi-token skill listings. The logs confirm that master records were successfully parsed and expanded into 60,796 distinct skill rows saved in optimized Parquet format.

Meet - ykb-xijh-nxa x Browsing HDFS x New Tab x + Ask Gemini

localhost:9870/explorer.html#/data/linkedin

Hadoop Overview Datanodes Datanode Volume Failures Snapshot Startup Progress Utilities -

Browse Directory

/data/linkedin Go

Show 25 entries Search:

	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
☐	drwxr-xr-x	hp	supergroup	0 B	May 30 18:02	0	0 B	processed
☐	drwxr-xr-x	hp	supergroup	0 B	May 30 18:05	0	0 B	raw
☐	drwxr-xr-x	hp	supergroup	0 B	May 30 18:02	0	0 B	results

Showing 1 to 3 of 3 entries

Previous 1 Next

Hadoop, 2023.

Activate Windows
Go to Settings to activate Windows.

92°F Sunny

Exploratory Data Analysis SQL Aggregations (Stage 2)

Stage 2 utilizes Spark SQL to execute complex analytical queries across the cluster nodes to uncover market patterns. The system logs validate that all 7 core analytical tables (covering geographic demand, salary splits, and remote work trends) were successfully generated and written.

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
-rw-r--r--	hp	supergroup	0 B	May 30 19:57	2	128 MB	_SUCCESS
drwxr-xr-x	hp	supergroup	0 B	May 30 19:55	0	0 B	sector=construction
drwxr-xr-x	hp	supergroup	0 B	May 30 19:55	0	0 B	sector=education
drwxr-xr-x	hp	supergroup	0 B	May 30 19:55	0	0 B	sector=energy
drwxr-xr-x	hp	supergroup	0 B	May 30 19:56	0	0 B	sector=finance
drwxr-xr-x	hp	supergroup	0 B	May 30 19:56	0	0 B	sector=healthcare
drwxr-xr-x	hp	supergroup	0 B	May 30 19:56	0	0 B	sector=hospitality
drwxr-xr-x	hp	supergroup	0 B	May 30 19:56	0	0 B	sector=legal
drwxr-xr-x	hp	supergroup	0 B	May 30 19:56	0	0 B	sector=logistics
drwxr-xr-x	hp	supergroup	0 B	May 30 19:56	0	0 B	sector=manufacturing
drwxr-xr-x	hp	supergroup	0 B	May 30 19:56	0	0 B	sector=marketing
drwxr-xr-x	hp	supergroup	0 B	May 30 19:56	0	0 B	sector=retail
drwxr-xr-x	hp	supergroup	0 B	May 30 19:56	0	0 B	sector=technology

Machine Learning Clustering Optimization (Stage 3)

To group similar job profiles, a distributed K-Means model is trained on TF-IDF text features and optimized using Silhouette evaluation scores across values of k from 4 to 12. The execution output explicitly confirms that $k=8$ was chosen as the mathematically optimal cluster configuration with a Silhouette score of 0.0824.

```
Administrator: Command Prompt - spark-submit --master yarn --deploy-mode cluster --num-executors 4 --executor-cores 3 --executor-memory 5g --driver-memory 2g --conf spark.executor.memoryOverhead=1g --conf spar...

[B] Sector monthly trend
    sector_monthly_trend
[C] Skill frequency
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k=10 silhouette=-0.0348
k=11 silhouette=-0.0421
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Browse Directory

Show 25 entries

Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	☐
☐	drwxr-xr-x	hp	supergroup	0 B	May 30 21:15	0	0 B	cluster_assignments	☐
☐	drwxr-xr-x	hp	supergroup	0 B	May 30 21:16	0	0 B	cluster_profiles	☐
☐	drwxr-xr-x	hp	supergroup	0 B	May 30 21:14	0	0 B	geo_heatmap	☐
☐	drwxr-xr-x	hp	supergroup	0 B	May 30 21:15	0	0 B	kmeans_k_selection	☐
☐	drwxr-xr-x	hp	supergroup	0 B	May 30 21:14	0	0 B	remote_trend	☐
☐	drwxr-xr-x	hp	supergroup	0 B	May 30 21:14	0	0 B	salary_distribution	☐
☐	drwxr-xr-x	hp	supergroup	0 B	May 30 21:14	0	0 B	sector_demand_geo	☐
☐	drwxr-xr-x	hp	supergroup	0 B	May 30 21:14	0	0 B	sector_monthly_trend	☐
☐	drwxr-xr-x	hp	supergroup	0 B	May 30 21:14	0	0 B	skill_freq_overall	☐
☐	drwxr-xr-x	hp	supergroup	0 B	May 30 21:14	0	0 B	skill_trend_yoy	☐

Data Visualization & Insights (Google Colab)

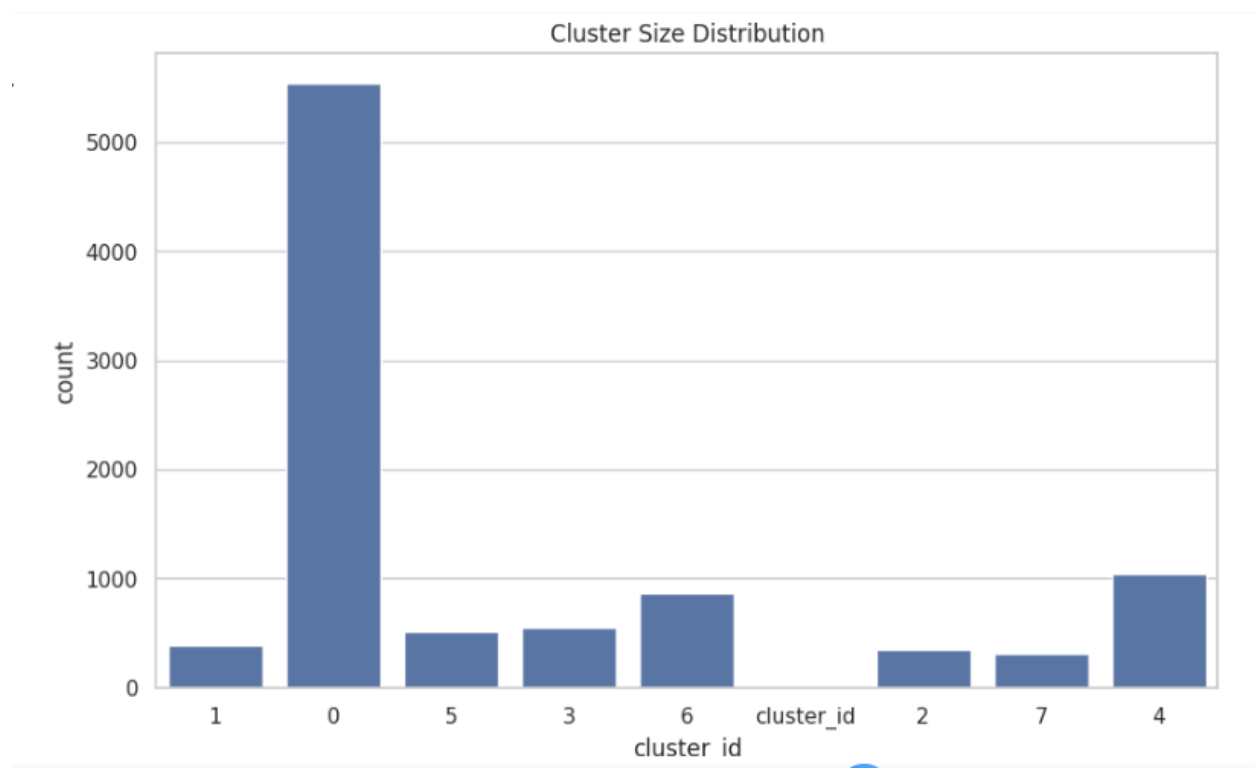
The finalized analytical result tables generated by the Spark cluster were imported into Google Colab for final-stage visualization and interpretation. Using Pandas, Matplotlib, and Seaborn, we generated high-quality plots, demand charts, and salary distribution graphs from the processed files to visually capture active labor market shifts and emerging skill patterns.

The processed LinkedIn job postings data, which was initially stored and analyzed using Hadoop HDFS and Apache Spark, was further explored through Exploratory Data Analysis (EDA). After preprocessing, feature engineering, aggregation, and clustering, the generated results were exported as CSV files and imported into Python for visualization and analysis. The EDA was conducted to gain insights into labor market trends by examining job cluster distributions, salary patterns, experience-level requirements, geographic demand, remote work adoption, sector-wise hiring trends, and skill demand.

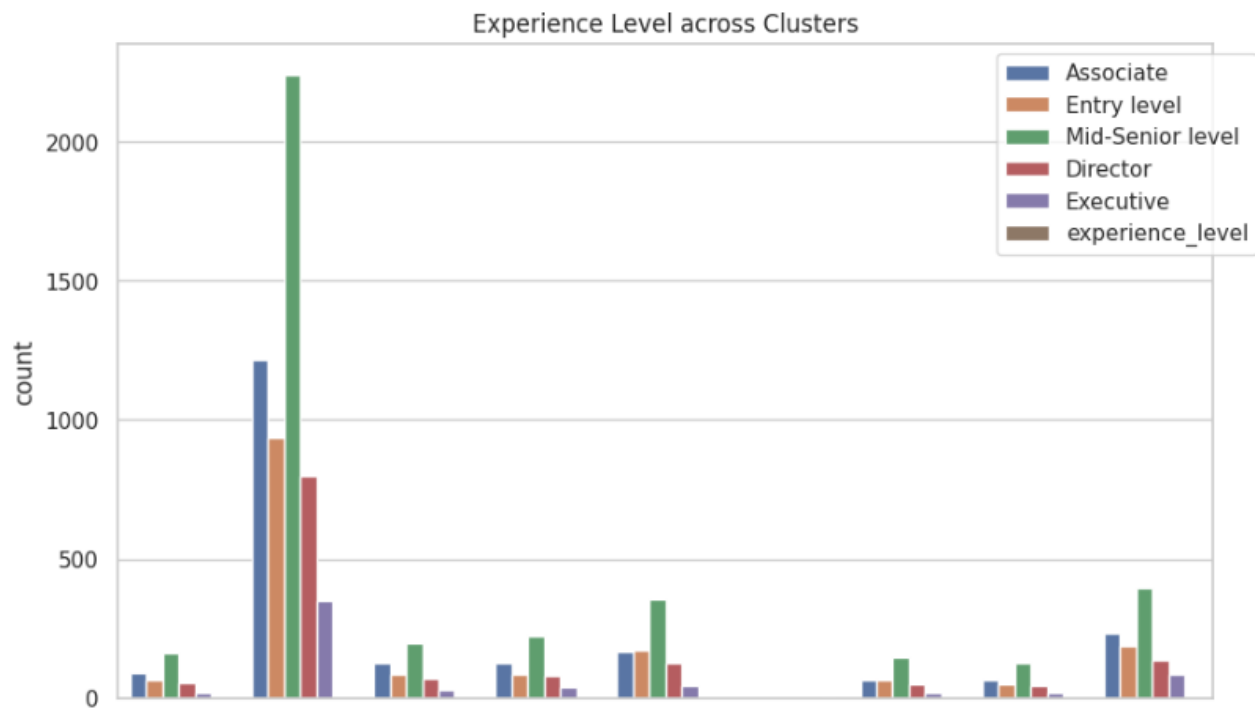
The results we recieved were in csv files which further isualised in colab

Results:

<https://drive.google.com/drive/folders/1HtGq29G8OczxUN1ZQJMaKibrQejvoRLP?usp=sharing>

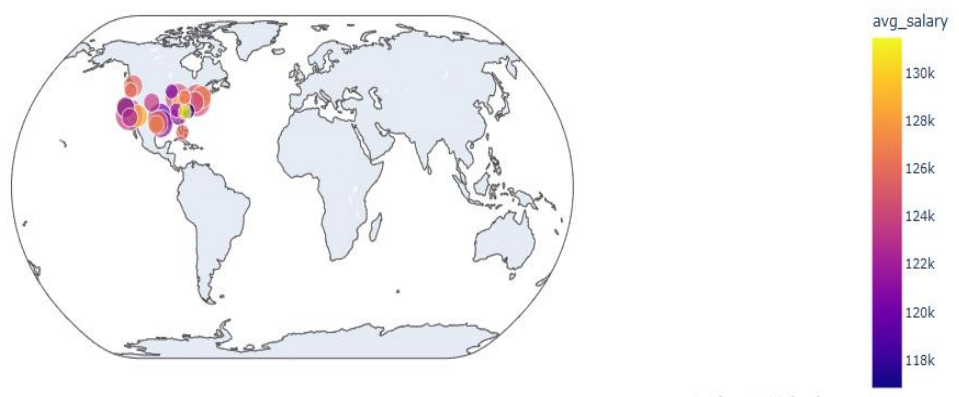


This visualization shows the number of job postings assigned to each cluster. It helps evaluate how jobs are distributed among the generated clusters and whether any cluster dominates the dataset.

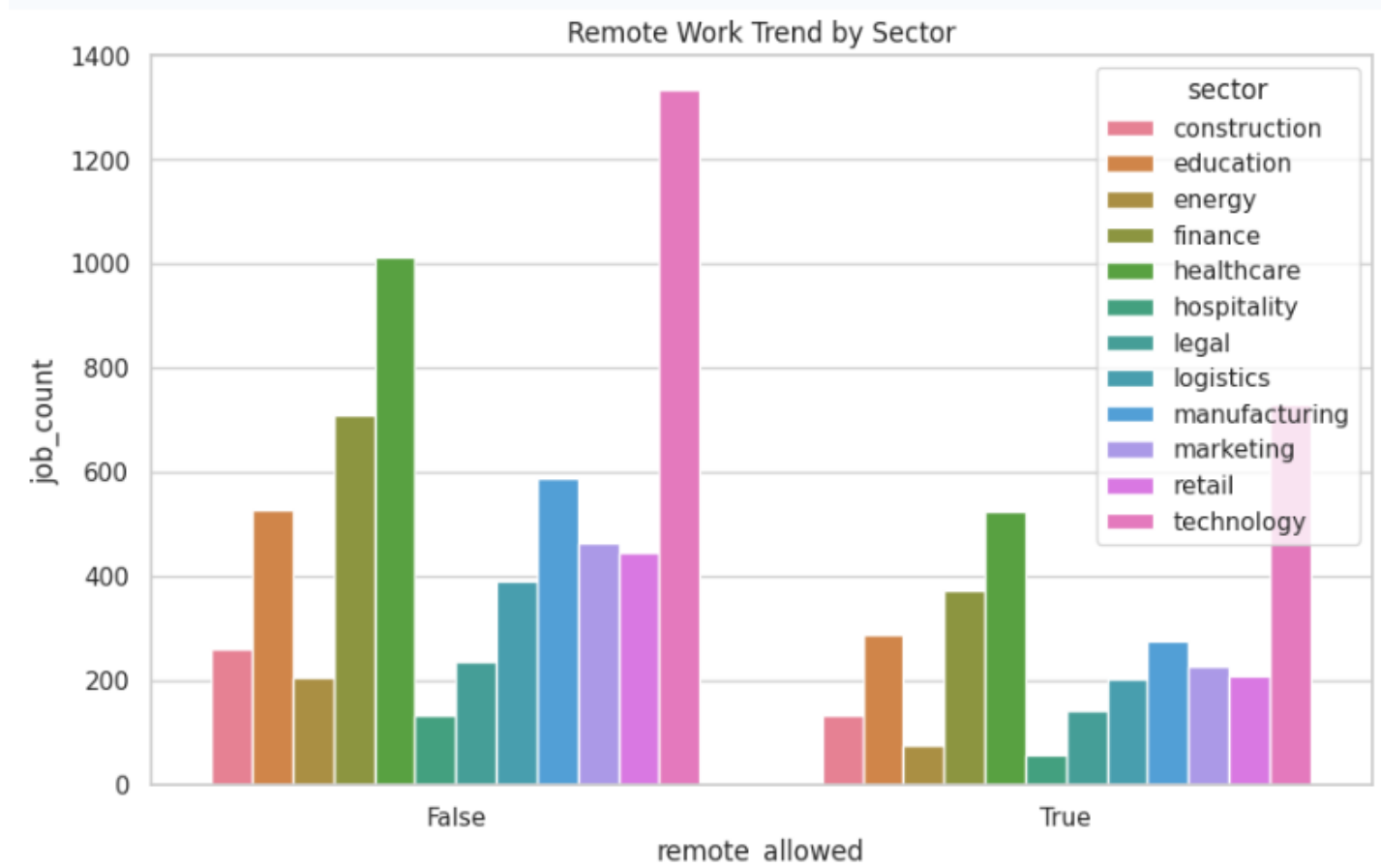
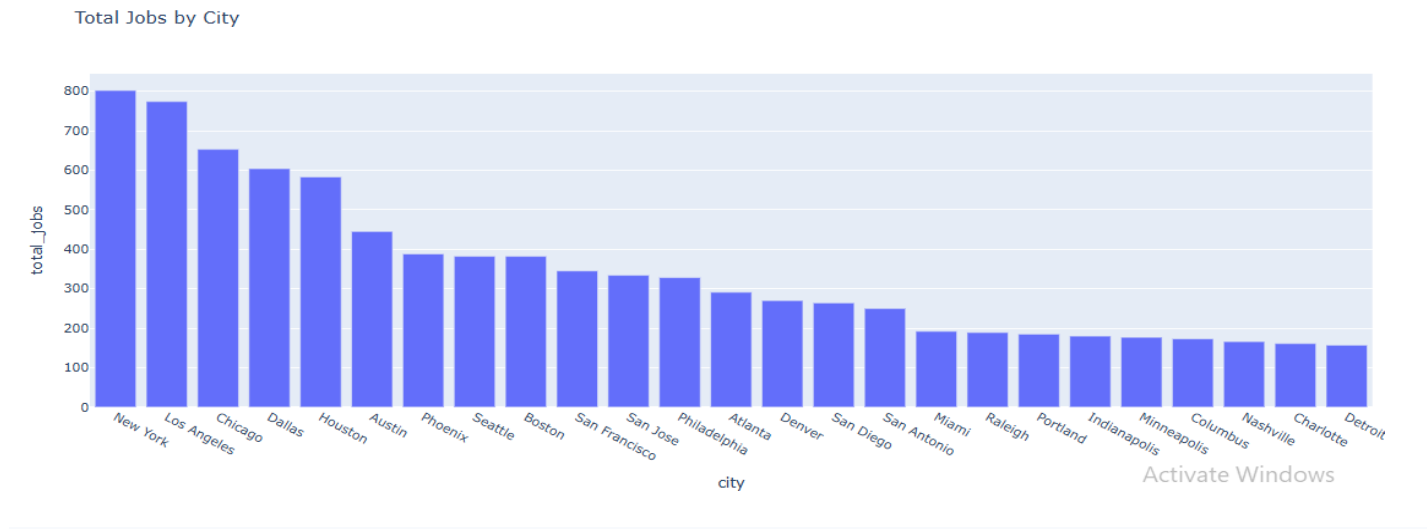


This chart illustrates how different experience levels are distributed within each cluster. It helps identify whether a cluster primarily contains entry-level, mid-level, or senior-level positions.

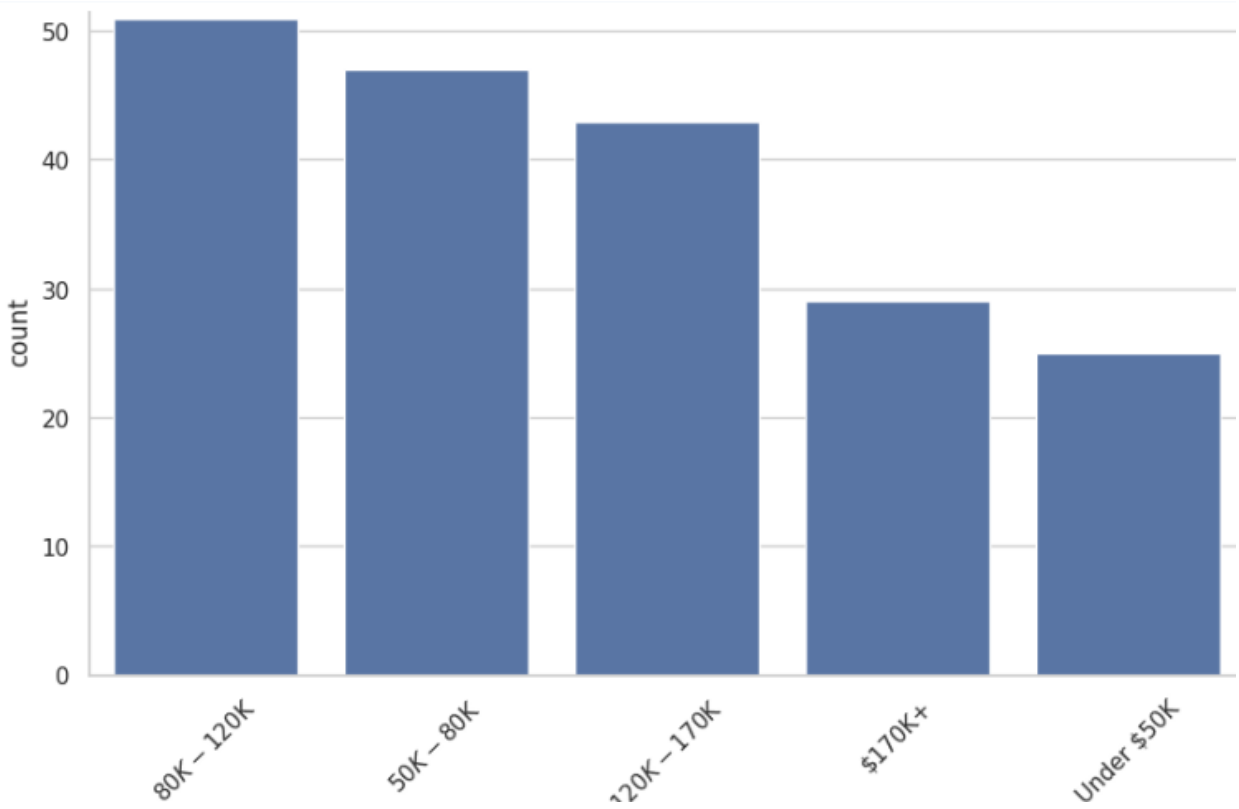
Job Density & Salary Heatmap



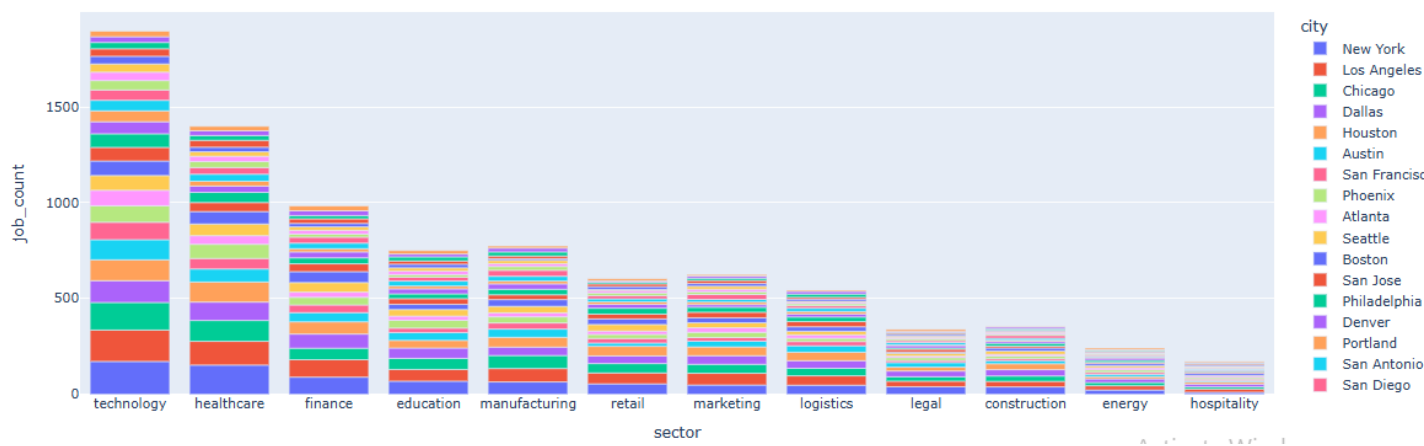
This interactive map visualizes geographic job demand and salary levels. Marker size represents job volume, while marker color represents average salary, allowing identification of important labor market regions.

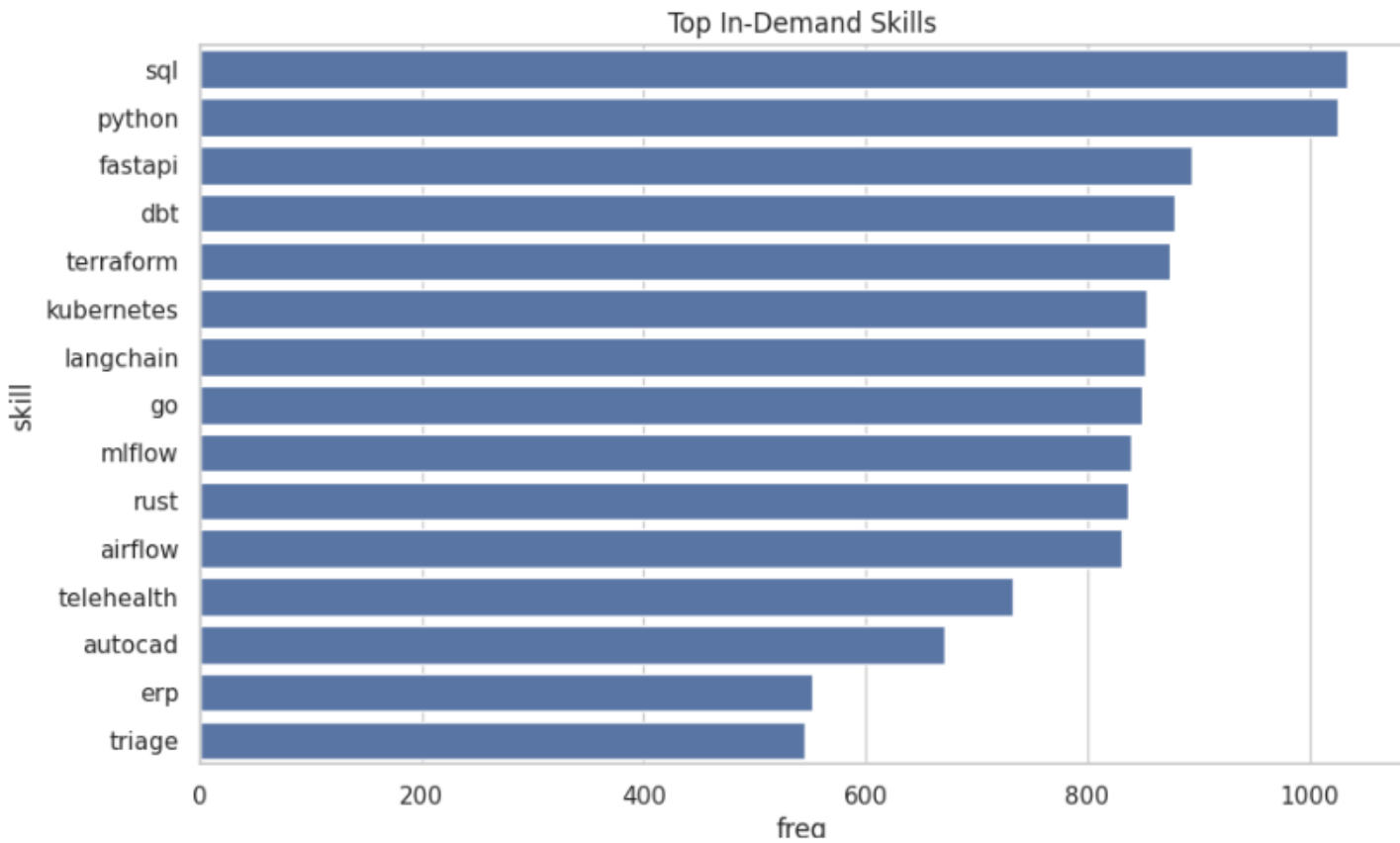


This analysis compares remote and non-remote job opportunities across industries, helping identify sectors that are more supportive of remote work arrangements.



Sector Demand by City





Results & Outcomes

The end-to-end big data analytics pipeline successfully executed across the 4-node on-premises cluster, completely processing the large-scale LinkedIn dataset to deliver real-time labor market insights. The distributed processing engine managed the intensive computation workloads seamlessly, bypassing traditional reporting delays and single-machine memory constraints. The system generated highly structured analytical files directly within the Hadoop Distributed File System (HDFS). These finalized output files were subsequently extracted and imported into Google Colab, where advanced visualization libraries were utilized to map out the exact economic indicators defined in the project objectives.

Through this collaborative cluster-to-notebook architecture, all three core expected outcomes were definitively achieved: regional hiring intensities were mapped via explicit sector-level metrics, temporal line plots successfully isolated emerging versus declining technical skills, and a fully optimized K-Means machine learning model exposed the underlying structural trends of modern professional roles.

<https://drive.google.com/drive/folders/1HtGq29G8OczxUN1ZQJMakibrQejvoRLP?usp=sharing>

Project Conclusion

The pipeline successfully delivers real-time economic insights, including geographic hiring maps, year-over-year skill trend arrays, and automated role clusters. By distributing the data across an on-premises cluster, the system proves that open-source Big Data tools can accurately detect sudden labor market shifts without reporting delays.

Attached links:

Dataset:

<http://kaggle.com/datasets/asaniczka/1-3m-linkedin-jobs-and-skills-2024>

Results:

<https://drive.google.com/drive/folders/1HtGq29G8OczxUN1ZQJMakibrQejvoRLP?usp=sharing>

https://drive.google.com/drive/folders/1fdzS_hF0khli0hbaU60wir_iqYUIXakY?usp=sharing

EDA:

https://colab.research.google.com/drive/1_gWIdjWSh7ynmlKXs-Jh5UTSM4J4vl7r?usp=sharing

Github Repository:

<https://github.com/k258001-code/linkedin-job-market-analysis>

